NeuroAgent

A benchmark and an agent stack for evidence-driven neurological diagnosis

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Outline

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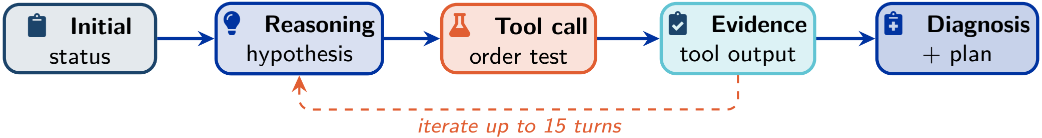
2. **Benchmark**
3. **Validation**
4. **Agent**
5. **Results**
6. **Outlook**

Motivation

Motivation Benchmark Validation Agent Results Outlook

Project goal

Build a benchmark *and* an agent pipeline that measure (and improve) evidence-driven clinical reasoning. Neurology is an ideal test bed (MRI, EEG, EMG, CSF, labs).



Realistic cases

500+ simulated neurology cases that mirror real practice: the agent uncovers diagnostic evidence step by step through tool calls, the way a clinician orders tests.

Multi-paradigm agents

A ladder of variants: vanilla LLM $\rightarrow$ tool use $\rightarrow$ SFT on golden traces $\rightarrow$ GRPO $\rightarrow$ graph-structured reasoning.

Expert validation

Every case reviewed by $\geq$3 clinicians on a custom platform. Inter-rater agreement = gold standard.

Medical LLM benchmarks reward trivia, not reasoning

How LLMs are evaluated today

- Datasets: MedQA Jin et al. (2021), MedMCQA Pal et al. (2022), USMLE-style sets
- Static multiple-choice questions
- Every relevant fact is in the stem
- Exactly one correct option
- Rewards recall + pattern matching

How clinicians actually work

- Start with a messy chief complaint
- *Decide* which tests to order
- Weigh cost, risk, time, stewardship
- Resist red herrings and anchoring
- Iterate evidence $\rightarrow$ hypothesis

The exam pre-solves the hardest step of diagnosis:
deciding *what evidence is worth looking at*.

The cost of measuring the wrong thing

Frontier LLMs reportedly match or beat clinicians on medical exams Singhal et al. (2023), yet real deployments tell a more complicated story.

What multiple-choice scoring cannot measure:

- **Tool selection:** did the agent order the right test at the right time?
- **Tool parameterisation:** with the right panel, sequence, or region?
- **Cost and stewardship:** did the agent order three MRIs when one would suffice?
- **Red-herring resistance:** did it chase the obvious finding past the contradicting one?
- **Reasoning trace:** can a clinician follow the logic from chief complaint to plan?

MedCaseReasoning Wu et al. (2025) makes the same point on reasoning fidelity; we extend it with *tool use*.

Two complementary contributions

NeuroBench

- **365** synthetic from per-condition YAML specs
- **151** seeded from real PMC reports (CC-BY 4.0) Wu et al. (2025); Zhao et al. (2022)
- Evidence moved out of the HPI (history of present illness) into tool outputs
- Rich ground truth: differential, optimal sequence, red herrings, contraindicated actions

NeuroAgent

ReAct Yao et al. (2023) tool-augmented agent under five progressively richer paradigms:

- Vanilla LLM (no tools): floor
- Tool-augmented (12 diagnostic tools)
- + SFT on golden trajectories
- + GRPO Shao et al. (2024) *(planned)*
- + graph-based structured reasoning *(planned)*

Closest prior work: AgentClinic

AgentClinic Schmidgall et al. (2024) pioneered agentic clinical evaluation; NeuroBench targets a complementary axis, evidence-driven tool use.

Dimension	AgentClinic (npj Dig. Med. 2026)	NeuroBench (ours)
Cases	122 (MedQA + NEJM); leakage risk	516 (synthetic + PMC-seeded); low leakage
Clinical scope	General USMLE breadth	Neurology, in depth
What it probes	Doctor–patient dialogue, history-taking	Tool selection on a stripped HPI
Tool model	Measurement agent returns values	12 typed tools, follow-up branching
Cost & stewardship	*None* (their stated future work)	Medicare-PFS cost per tool call
Scored on	Diagnosis label only	Dx + optimal / contraindicated actions
Replay	LLM agents: non-deterministic	Static cache: deterministic, GPU-free
Beyond evaluation	Benchmark only	Benchmark + SFT → GRPO training

Complementary, not competing: dialogue-driven history-taking (AgentClinic) vs. evidence-driven tool use (ours).

Benchmark

Motivation — **Benchmark** — Validation — Agent — Results — Outlook

NeuroBench at a glance

Cases	Tools	Follow-ups
516	12	3,307
20 conditions	Pydantic-typed outputs	pre-cached outputs

What it is. 500+ simulated neurology patients for evaluating tool-augmented LLM agents on clinical reasoning, not pattern recognition.

Distinctive properties.

- 20 conditions spanning neuromuscular, vascular, demyelinating, dementia, and more
- 12-tool schema with Medicare-PFS cost tracking
- 3 difficulty tiers; mix of synthetic and PMC-seeded

Two generation pipelines, one schema

Synthetic pipeline $n = 365$

Per-condition YAML spec defines disease, expected tool outputs, and reasoning rubric. LLM receives:

1. Pydantic JSON schema (`NeuroBenchCase`)
2. Condition YAML
3. Difficulty tier + encounter type

Real-PMC-seeded pipeline $n = 151$

Seed: a published PMC case report (Wu et al. (2025)).

The LLM is instructed to:

1. Preserve the clinical trajectory
2. Strip diagnostic results out of the HPI
3. Move them into structured tool outputs
4. Add red herrings at higher difficulties

Both pipelines emit the same schema. Case IDs starting with `R` are PMC-seeded (e.g. `ALS-RS11`); others are synthetic.

Anatomy of a case: building the patient picture

What the agent sees at turn 1

Demographics	Chief complaint	HPI narrative	Vitals	Neuro. exam
Past medical hx	Medications	Allergies	Family hx	Social hx

Revealed only through tool calls: the 12-tool catalogue

analyze_brain_mri	analyze_eeg	analyze_ecg	interpret_labs
analyze_csf	order_ct_scan	order_echocardiogram	order_cardiac_monitoring
order_advanced_imaging	order_specialized_test	search_medical_literature	check_drug_interactions

Together these compose the complete patient; the agent must call the right tools to fill in the orange blocks.

Case study: ALS-M01 at a glance

Field	Value
Case ID / condition	`ALS-M01`: ALS (bulbar-onset)
Difficulty / encounter	moderate (concurrent cervical spondylosis), outpatient
Patient	62 y female, right-handed, retired teacher, BMI 23.1
Chief complaint	Progressive dysarthria + dysphagia + limb weakness over 8 months
Ground-truth Dx	ALS bulbar-onset, moderate stage; ICD-10 G12.21
Initial tool outputs	MRI brain + cervical spine, labs, EMG/NCS
Follow-ups ($n = 6$)	PFTs, genetic panel, CSF NfL, repeat EMG, MDC referral, riluzole check

Pedagogic point: tests whether the agent commits to neurosurgical referral for the cervical spondylosis, or recognises that bulbar findings plus supra-cervical EMG denervation cannot be explained by cervical disease.

What the agent sees at turn 1

Output of `format_patient_info()`

```
Patient: 62 yo female. CC: progressive dysarthria, dysphagia, and
  bilateral limb weakness over 8 months.
HPI: 62-yo right-handed woman with 8-month progressive dysarthria,
  dysphagia, and limb weakness. PCP ordered a cervical MRI showing
  moderate spondylosis; cervical myelopathy raised as etiology...
PMH: HTN; hypothyroidism; osteoporosis; remote right CTS released 5y ago
Exam:
  Motor: deltoids 4/5, biceps 4+/5, hand 4-/5 R, 3+/5 L. Visible
    fasciculations bilateral UE proximal muscles.
  Reflexes: hyperreflexia 3+; plantars extensor (Babinski +);
    Hoffman sign + bilaterally.
  Add'l: fasciculations in tongue, deltoids, left thenar...
Vitals: BP 128/78, HR 72, T 36.8 C, RR 14, SpO2 97%

```

No MRI report, no labs, no EMG, no CSF; the agent decides what to order.

Hidden inside the case: what tools return

`analyze_brain_mri` (initial)

Bilateral CST T2 hyperintensity in posterior limb of internal capsule. Cervical spondylosis, C5–C6 stenosis, *no cord signal change*. Confidence: 0.0; impression reads "clinical correlation recommended."

`interpret_labs` (NM panel)

CK **748 U/L** (H), Aldolase 8.2 (H, mild). TSH, B12, ESR, CRP normal. Anti-GM1 neg. SPEP no M-spike. AR-CAG 22 (*normal*).

`order_specialized_test` (EMG/NCS)

Active denervation + chronic reinnervation across **bulbar, cervical, thoracic**; early lumbosacral changes. Sensory NCS normal. No conduction block.
El Escorial revised: clinically definite ALS.

`analyze_csf` (follow-up)

OP 14 cmH_2O; WBC 2; protein 58 (mildly elevated). NfL **8,420** pg/mL (ref $<1{,}000$). Oligoclonal bands absent. Cytology negative.

Each output is a Pydantic model with full panel structure and per-test abnormality flags.

Off-pathway tool calls stay in character

A case pre-caches outputs only for its diagnostic pathway, yet the agent may call any of the 12 tools. A second fallback tier answers the rest, so the simulation never returns a "no data available" error.

What an off-pathway tool returns

- **Non-contributory normal** (87%): a properly structured normal report for that modality
- **Incidental finding** (13%): abnormal but not the diagnosis, such as chronic small-vessel change, a benign cyst, or trace valvular regurgitation

Incidental rate calibrated to the imaging-incidentaloma literature (15–30% of scans).

Why it matters

Over-ordering is no longer a free dead-end: it costs money, burns turns, and forces the agent to dismiss red herrings it created itself.

2,813 fallback outputs, drawn deterministically per (case, tool) and flagged for the evaluator, so over-ordering is always scored.

Ground truth grades the *reasoning path*, not just the answer

Optimal action sequence (excerpt)

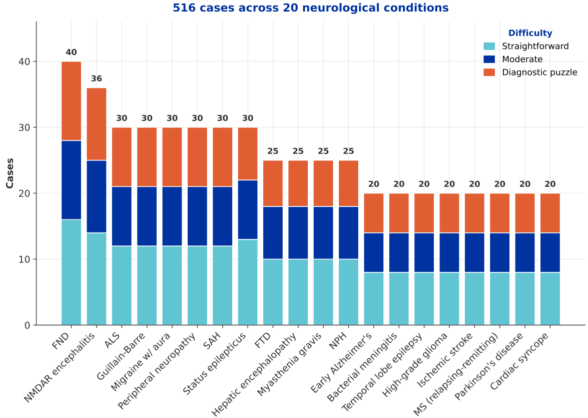
1. MRI brain + cervical spine — *required*
2. Labs: CK, anti-GM1, SPEP, TSH — *required*
3. EMG/NCS across 4 regions — *required*
4. Pulmonary function (FVC, SNIP) — *required*
5. ALS genetic panel — *acceptable*
6. Riluzole + ALS MDC referral — *required*

Red herrings encoded in the case

- **C5–C6 stenosis** pulls toward neurosurgery. Correct: no cord signal change; bulbar findings unexplained by cervical pathology.
- **Remote right CTS** suggests entrapment for hand weakness. Correct: EMG denervates non-median muscles bilaterally.

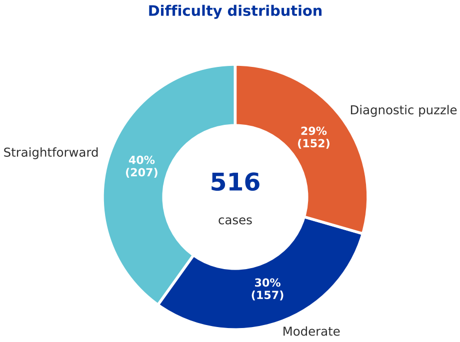
Each step stores `expected_finding` and `tool_parameters`, so the evaluator grades not just *whether* the right test was ordered but *how* it was parameterised.

20 conditions across neurology subspecialties



Coverage spans neuromuscular (ALS, MG, GBS, PN), epilepsy/status, vascular (stroke, SAH), dementia (AD, FTD, NPH), demyelinating (MS, NMDAR), infectious/metabolic, headache, neuro-oncology, movement, and FND.

Difficulty calibration



Straightforward (40%)

Classic textbook presentation. An intern should get this right.

Moderate (30%)

Atypical features or subtle findings. 1–2 red herrings (incidental labs, borderline imaging, confounding meds).

Diagnostic puzzle (30%)

Initial presentation mimics a different condition. Key diagnostic clue buried in follow-up results.

Validation

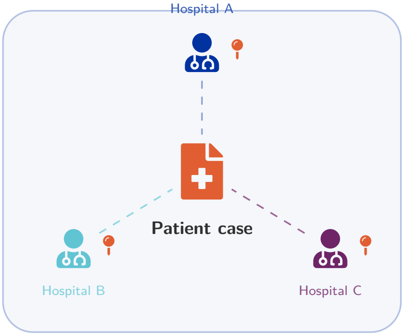
Motivation — Benchmark — **Validation** — Agent — Results — Outlook

Synthetic data alone is not enough

The benchmark is only as credible as the cases inside it. LLM-generated trajectories can be:

- Internally inconsistent (lab values that contradict the diagnosis)
- Clinically implausible (mis-staged disease, wrong-age presentation)
- Mis-keyed red herrings that aren't actually distracting

Our approach: every case reviewed by $\geq$**3 independent clinicians** on a dedicated platform, with field-level annotations and inter-rater agreement tracking.



The review platform: Overview tab

What it is

Separate FastAPI + Vite app, port 8889, gated by `X-Reviewer-Code` header. File-based persistence under `data/review/annotations/`. Reuses the same `NeuroBenchCase` schema.

Four tabs

- **Overview:** per-reviewer progress dashboard
- **Cases:** 516 cases, field-level annotation
- **Methodology:** public showcase
- **Admin:** aggregate agreement views

APP SCREENSHOT

Overview tab: reviewer greeting, per-condition progress bars, recent activity feed, status pie (pending / in-progress / approved / needs-changes).

Annotation workflow: field-level, gesture-driven

Status flow (per reviewer, per case)

`pending` → `in_progress` *(auto on first annotation)* → `needs_changes` / `approved`

Annotation gesture

Hover any `AnnotatableField` → Comment pill in margin → Radix Popover with severity (note / issue / error), textarea, and Cmd+Enter save. Annotated fields get a left border coloured by highest severity.

Filesystem isolation

One JSON file per (reviewer, case) pair, so a reviewer endpoint cannot return another reviewer's data.

APP SCREENSHOT

Cases tab: annotated MRI report with a margin Comment pill open into a Radix Popover with severity (note / issue / error), textarea, and Cmd+Enter save.

Admin tab: aggregate views over all reviewers

Four aggregate views

- **Inter-rater agreement:** per-case verdict + Cohen's κ
- **Reviewer progress:** cases over time
- **Field hotspots:** most *error*-tagged fields
- **Side-by-side diff:** compare 3 reviewers on the same case

Output: a trusted dataset

Cases with $\geq$3 *approved* verdicts and high κ graduate into the locked benchmark.

APP SCREENSHOT

Admin dashboard: inter-rater agreement heatmap (rows = cases, cols = reviewers) with Cohen's κ card; reviewer-progress sparkline; field-hotspot bar chart.

Agent

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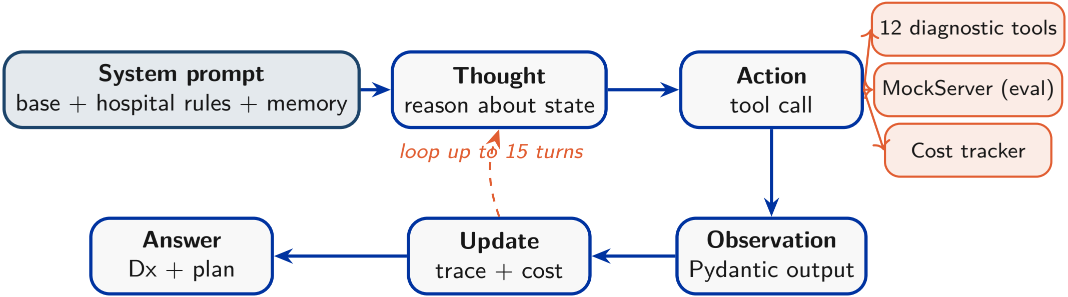
NeuroAgent architecture: ReAct loop up to 15 turns



Built on ReAct Yao et al. (2023); tool-augmentation pattern from Toolformer Schick et al. (2023). In evaluation mode, `MockServer` replays pre-generated tool outputs, with no GPU needed.

The 12 diagnostic tools

Tool	Returns	Cases
`interpret_labs`	lab panel, reference ranges, abnormality flags	516
`analyze_brain_mri`	MRI report, region-tagged abnormalities	376
`analyze_eeg`	EEG: background, epileptiform features, focality	362
`analyze_ecg`	ECG: rhythm, intervals, ischemia, dysrhythmia	314
`analyze_csf`	CSF: cells, protein, glucose, NfL, OCB, cytology	302
`search_medical_literature`	curated guideline / paper snippets	216
`check_drug_interactions`	interaction profile + contraindications	216
`order_specialized_test`	EMG/NCS, biopsy, autoimmune panels	169
`order_ct_scan`	CT: non-contrast / contrast, head / spine	139
`order_advanced_imaging`	MRA / MRV / CTA / PET / DAT-SPECT	67
`order_echocardiogram`	TTE / TEE: chambers, valves, EF, thrombus	57
`order_cardiac_monitoring`	telemetry / Holter / loop recorder summary	57

All outputs are Pydantic models (`.model_dump()`); each tool has a Medicare-PFS reference cost logged by `CostTracker`.

Hospital protocols and patient memory

Hospital protocols

YAML files at `config/hospital_rules/{hospital}/`, one file per pathway (stroke, status epilepticus, bacterial meningitis, ...).

At session start the system prompt is augmented with the relevant pathway rules.

Real hospitals *differ* on which panel, which contrast, when to escalate, and the benchmark must reflect that.

Patient memory

Per-patient long-term context: prior visits, chronic conditions, baseline cognition, advance directives. Reasoning should improve when the agent remembers a 6-month gradient instead of treating every encounter as cold.

System prompt at runtime

```
base_prompt
  + hospital_rules(pathway)
  + patient_memory(patient_id)
```

Serving stack and hardware

Models

Four open-weights backbones via **vLLM** Kwon et al. (2023) on **A100-40 GB**:

- Qwen3.5 with `qwen3` reasoning parser plus `qwen3_coder` tool parser
- MedGemma with `hermes` tool parser
- Two further variants for ablation

Mac dev: Ollama fallback. Defaults: `T=1.0`, `top_p=0.95`, `pp=1.5`, `max=8192`.

Client and orchestration

`llm/client.py` strips <think> tags from Qwen and parses OpenAI-style tool calls. FastAPI port 8888 serves REST + SSE streaming. Loading via `POST /api/v1/models/{key}/load` (SSE progress).

Trace replay

Every run is auto-saved to `data/traces/` so we can re-score without GPU time; crucial when iterating on metrics.

Training: golden trajectories → SFT

Building the SFT corpus

1. Take the case's ground-truth optimal action sequence
2. Choreograph a teacher rollout: tool call → cached output → short rationale → next call
3. Verify the trace terminates in the correct diagnosis and respects critical / contraindicated actions
4. Filter by clinician-validated cases only (κ-filtered)
5. Emit (`system`, `user`, `assistant`) chat tuples

Targets

Smaller deployable backbones (7–14 B). Goal: match larger frontier baselines on tool selection and diagnostic accuracy at a fraction of the inference cost.

What SFT does *not* cover

Teaches imitation of an optimal trajectory, not recovery from divergence; that's why we want RL on top.

Fine-tuning method: QLoRA on Qwen3.5-9B

What actually trains

- Base `Qwen3.5-9B` frozen in 4-bit NF4 (BitsAndBytes, double quant, bf16 compute)
- LoRA adapters `r=64`, α=128 on *all* attention + MLP projections
- Only adapters update ($\sim$tens of MB), merged for deployment; one recipe for SFT *and* RL

Why QLoRA: a single A100-40 GB

- Full fine-tuning needs $\sim 4\times$ the weights for optimiser states — multi-GPU only
- 4-bit base shrinks 18 GB→5 GB, leaving room for 6144-token ReAct traces
- NF4 + double quant near-lossless: matches 16-bit LoRA quality Dettmers et al. (2023)

Alternatives considered

Full fine-tuning (multi-GPU only), **plain 16-bit LoRA** (no memory headroom), **DoRA / LoftQ** (quality upgrade, future ablation), and **prefix / prompt tuning** (too weak for tool use).

What's next: GRPO and graph-structured reasoning

GRPO: relative reasoning RL)

Sample k trajectories per case, score each, optimise with rewards relative to the group mean, with no value network.

Why it fits:

- Reasoning quality is a relative signal across rollouts on the same case
- Our metrics (Dx acc, tool match, cost, red-herring) are already group-comparable

Graph-based structured reasoning

Replace free-text `Thought` with a typed *clinical reasoning graph*:
nodes $\in$ {symptom, exam finding, hypothesis, evidence, action, conclusion}; edges $\in$ {supports, contradicts, triggers, depends-on}.

Why:

- Constrains hallucination at generation
- Makes reasoning auditable by clinicians
- Maps onto our optimal action sequence

Both are research directions, not yet implemented. The benchmark is designed so progress on either is directly visible in the metrics.

Results

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How we evaluate: six complementary metrics

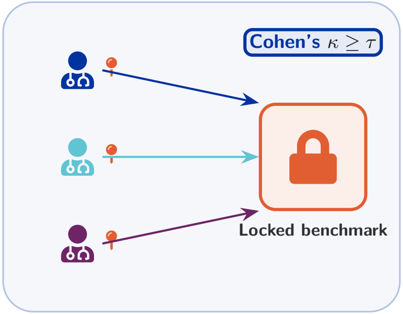
Metric	What it captures
Diagnostic accuracy	ICD-10 match against ground truth (exact + group-level)
Tool selection	precision / recall of called tools vs. optimal action sequence
Tool parameterisation	right panel / sequence / region, scored per step
Cost efficiency	sum of Medicare-PFS rates over all tool calls
Red-herring resistance	did the agent commit to a contraindicated / red-herring action?
Clinician score	aggregated verdict from the validation platform

Every metric runs against the same evaluator (`evaluation/runner.py`) with `format_patient_info()` as the single source of truth, so numbers across variants are directly comparable.

Clinician validation as the gold standard

From reviewer pool to score

1. Each case reviewed by $\geq$3 clinicians
2. Per-field annotations (note / issue / error) and a verdict (`approved` / `needs_changes`) stored on the platform
3. Admin computes Cohen's κ per field type
4. Cases with κ above threshold + consistent verdicts enter the *locked* benchmark
5. Agent runs are scored only on locked cases



Current results: model comparison on NeuroBench

Variant	Dx %	Tool F1	Param %	Cost $	Red-h %
Vanilla LLM (Qwen3.5)	*tbd*	*tbd*	*tbd*	*tbd*	*tbd*
+ Tools (ReAct)	*tbd*	*tbd*	*tbd*	*tbd*	*tbd*
+ Tools + SFT (golden)	*tbd*	*tbd*	*tbd*	*tbd*	*tbd*
MedGemma + Tools	*tbd*	*tbd*	*tbd*	*tbd*	*tbd*
Frontier reference	*tbd*	*tbd*	*tbd*	*tbd*	*tbd*
+ *GRPO (planned)*	–	–	–	–	–
+ *Graph reasoning (planned)*	–	–	–	–	–

Numbers from the latest full evaluation sweep. Dx = ICD-group match; Tool F1 vs. optimal sequence; Param = correct parameters; Cost in Medicare-PFS $/case.

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Outlook

Motivation Benchmark Validation Agent Results **Outlook**

Recap and roadmap

What we've shown

- Medical LLM eval needs to move past Q&A
- NeuroBench: 500+ cases, 20 conditions, 12 tools, evidence revealed via tool calls
- Custom platform for triple-blind clinician validation with inter-rater agreement
- NeuroAgent: ReAct loop, hospital protocols, memory, cost tracking, vLLM on A100s
- Variant ladder all measured on the same metrics

Next steps (paper milestones)

- Finish clinician validation on all 516 cases
- Lock benchmark + freeze metrics
- Full sweep across the variant ladder
- Implement GRPO; prototype graph reasoning
- Draft for **Nature Machine Intelligence**

Open questions

Metric weighting? Reviewer pool size? Conditions to expand?

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